Sheet (5)

(1) Write a Python program in which the user is promoted to enter a number and find its cube using function.

(2) Write a Python program, which prompt the user to enter two numbers and use functions to add, subtract and multiply these numbers.

(3) Write a Python program, which prompt the user to enter two numbers and use a function to swap the values the two numbers.

(4) Write a Python program that prompts the user to enter an integer and use a function, based on the *trial division method*, to check whether it is prime or not.

(5) Write a Python program to find the prime integers between a lower and upper value. Prompt the user to enter these values and use a function to find the prime number.

(6) Write a Python program, which prompt the user to enter an integer and use a function to calculate the factorial of that integer.

(7) Write a Python program, using function, to find the sum of n-terms of Arithmetic Progression Series. This series is expressed as;

$$a\,.\,a+d.\,a+2d\,.\,a+3d\,.\;\ldots$$

(8) Write a Python program, using function, to find the sum of geometric progression series. This series is expressed as;

$$a\,.\,ar.\,ar^2\,.\,ar^3\,.\quad\ldots$$

(9) Write a Python program, using a function, to generate a random 4-digit number and find the sum of the digits of that number.

(10) Write a Python program to calculate and print the sum of the following series. Use a function to calculate the sum of the series.

$$x+\frac{x^3}{3!}+\frac{x^5}{5!}+\;\ldots\;+\frac{x^n}{n!}$$

(11) Write a Python program to display the n terms of a harmonic series and calculate their sum using functions. The sum of first m terms of harmonic series is expressed as follows;

$$\sum_{n=1}^{n=m} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ \ldots + \frac{1}{m}$$

(12) Write a Python function with four arguments initialized with zero as default values for these arguments. This function is called from the main program to calculate the sum of arguments with assigned values in the calling statement.